is faster than the sound we hear through the air (gaseous medium). This implies the velocity of sound is different in different media.

In this section, we shall derive the velocity of waves in two different cases:

1. The velocity of a transverse waves along a stretched string.
2. The velocity of a longitudinal waves in an elastic medium.

11.3.1 Velocity of transverse waves in a stretched string

Let us compute the velocity of transverse travelling waves on a string. When a jerk is given at one end (left end) of the rope, the wave pulses move towards right end with a velocity v with respect to an observer who is at rest frame.

Consider an elemental segment in the string as shown in the Figure 11.15. Let A and B be two points on the string at an instant of time. Let *dl* and *dm* be the length and mass of the elemental string, respectively. By definition, linear mass density, μ is

$$\mu = \frac{dm}{dl} \tag{11.5}$$

$$dm = \mu\, dl \tag{11.6}$$

The elemental string AB has a curvature which looks like an arc of a circle with centre at O, radius R and the arc subtending an angle θ at the origin O as shown in Figure 11.15(b). The angle θ can be written in terms of arc length and radius as $\theta = \frac{dl}{R}$. The centripetal acceleration supplied by the tension in the string is

$$a_{cp} = \frac{v^2}{R} \tag{11.7}$$

Then, centripetal force is

$$F_{cp} = \frac{(dm)v^2}{R} \tag{11.8}$$

From eqn 11.6,

$$\frac{(dm)v^2}{R} = \frac{\mu v^2\, dl}{R} \tag{11.9}$$

The tension T acts along the tangent of the elemental segment of the string at A and B. Since the arc length is very small, variation

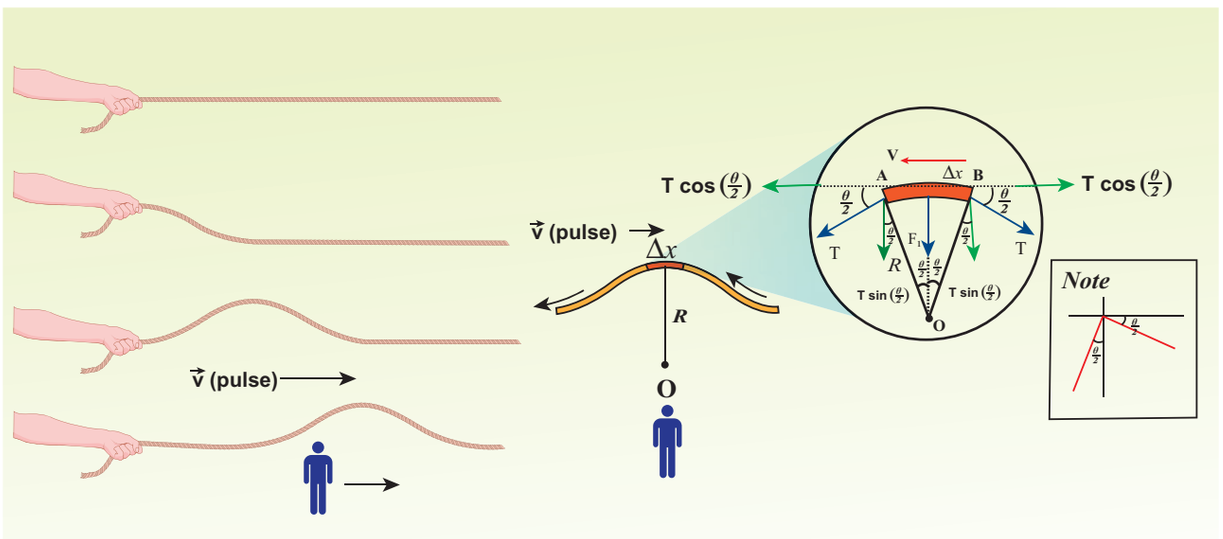


Figure 11.15 Elemental segment in a stretched string is zoomed and the pulse seen from an observer frame who moves with velocity v.

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